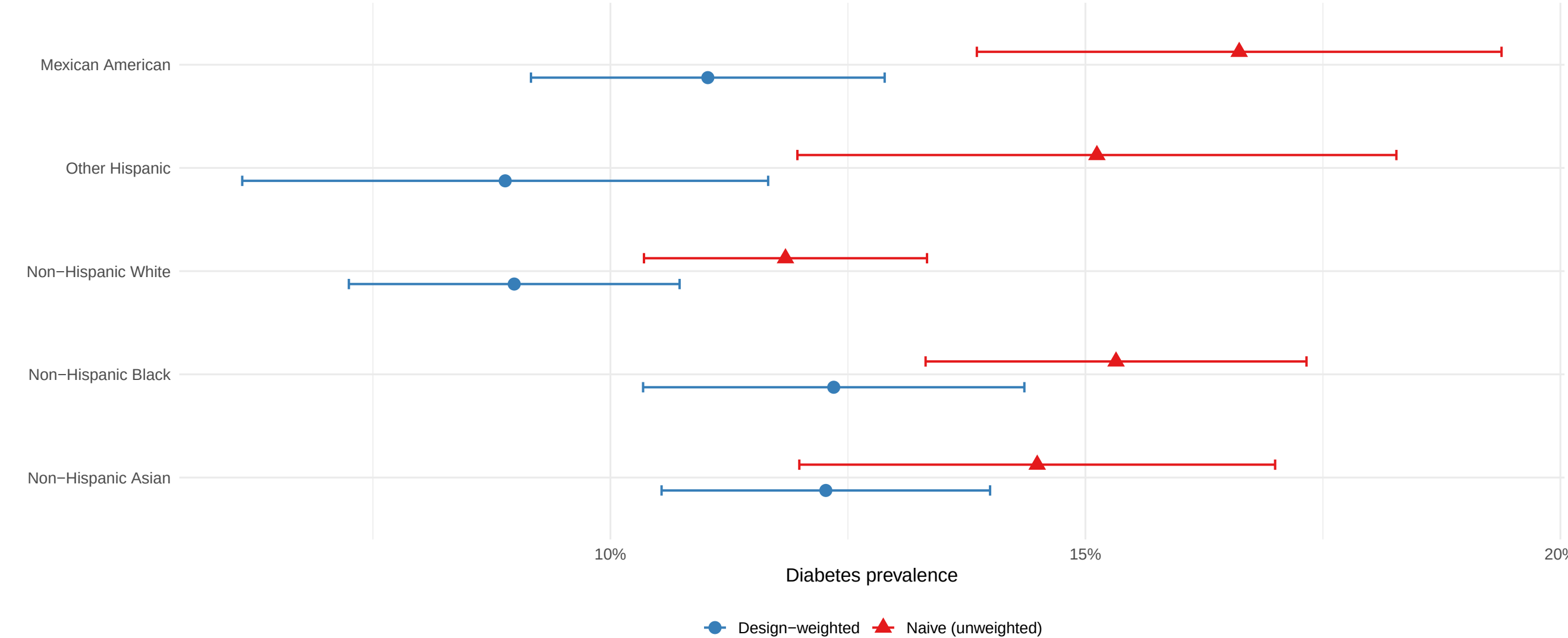


NHANES 2017–2018: Design Effects in Health Equity Research

National Health and Nutrition Examination Survey (n = 5,265 adults 20+). Complex design: oversampled minorities, stratified, clustered.
Design: `svydesign(id=~SDMVPSU, strata=~SDMVSTRA, weights=~WTMEC2YR, nest=TRUE)`. `survey.lonely.psu = 'adjust'`.

A. Diabetes Prevalence by Race: Does the Survey Design Matter?

NHANES 2017–2018 adults 20+. Diabetes defined as HbA1c $\geq 6.5\%$.
NHANES oversamples minorities. Naive estimates ignore this, distorting health disparity conclusions.



B. Standard Errors: Complex Design vs Simple Random Sample

NHANES clustering inflates SEs. $DEFF > 1$ means the real SE is larger than naive calculations assume.
Ignoring the design produces confidence intervals that are too narrow -- leading to false precision in health estimates.

